

RACER-GT: A Design-Based Statistical Framework for the Reliable Construction of Long-Horizon Google Trends Time Series

Randomized Acquisition, Global Overlap Calibration,
Generalizability
Decomposition, and Covariance-Adjusted Consensus

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Abstract

Google Trends (GT) is widely used in finance and economics as a high-frequency proxy for investor attention, information demand, and sentiment. Yet the web interface rescales every request to a 0–100 index, a long daily history must be assembled from several short query windows, and repeating an identical query can return different numbers. A researcher who downloads a series once, stitches the windows sequentially, and feeds the result into a financial model therefore inherits four distinct problems at once: segment-scale error, dependence across repeated retrievals, zero inflation and rounding at low search volume, and attenuation bias in any downstream regression.

This paper develops RACER-GT (Randomized Acquisition, Calibration, Error Decomposition, and Reliability for Google Trends). The framework does not claim to recover Google’s absolute search counts. It instead defines two estimands that *are* identified—the expected GT return under a fixed query and a fixed retrieval design, and the latent relative search signal on a common

scale—and then estimates them through eight preregistered stages: (i) a balanced, protocol-locked repeated-retrieval experiment; (ii) global weighted least squares on the overlap graph, which uses every usable overlap simultaneously instead of accumulating error through sequential stitching; (iii) exhaustive retention of exact duplicates together with a residual-based near-duplicate dependence graph; (iv) a three-facet Generalizability Theory decomposition over historical date, collection day, and collection stream; (v) a design-based stratified reference estimator and a covariance-adjusted minimum-variance consensus; (vi) quadratic-programming temporal benchmarking against weekly or monthly aggregates; (vii) separate detection, level, and daily-innovation reliability with a locked acceptance decision tree; and (viii) propagation of the resulting measurement uncertainty into reliability-corrected OLS and SIMEX.

We state the identification conditions explicitly and prove conditional unbiasedness and minimum-variance results under them. In twenty controlled Monte Carlo replications, RACER-GT attains a mean RMSE of 3.5265 against a known latent truth, compared with 7.2305 for a single pull, 4.1710 for the cross-pull median, and 3.7253 for the simple cross-pull mean. These results support the claim that the framework reduces reconstruction error under the stated data-generating process; they are not a universal optimality guarantee across keywords, regions, or GT versions.

Keywords: Google Trends; investor attention; measurement error; experimental design; generalizability theory; graph calibration; generalized least squares; temporal benchmarking; SIMEX.

JEL classification: C18, C43, C81, G14.

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1 Introduction

Search data have become a standard empirical input in finance. Da et al. (2011) showed that Google search frequency for ticker symbols captures retail investor attention in a way that pre-existing proxies do not; Preis et al. (2013) documented associations between search volume for finance-related terms and subsequent market movements; Choi and Varian (2012) established the broader “predicting the present” agenda. A large applied literature now treats a downloaded GT series as if it were an observed variable measured without error.

That treatment is difficult to defend once the retrieval mechanism is examined. Three features of the GT interface interact badly with long-horizon daily research designs.

Per-request renormalization. Every GT response is rescaled so that the maximum value within the requested window equals 100. The returned numbers are therefore not comparable across requests with different windows. Two requests that differ only in their end date produce two different scales for the same calendar day.

Window-length quantization. Daily granularity is returned only for sufficiently short windows. A multi-year daily history must be requested as a sequence of overlapping chunks and reassembled. The conventional reassembly is sequential: chunk 2 is rescaled onto chunk 1, chunk 3 onto the rescaled chunk 2, and so on. Scale errors then compound multiplicatively along the chain, and the final segment inherits the accumulated error of every preceding join.

Non-determinism. Repeating an identical query on a different day, or from a different browsing environment, can return different values. Eichenauer et al. (2022) document this instability directly and propose averaging across repeated samples. Averaging is a substantial improvement over a single download, but it raises a further question that the applied literature has largely left open: repeated retrievals are *not* independent samples, so what exactly does their average estimate, and how much information do N retrievals actually contain?

1.1 What this paper does

We treat data acquisition itself as a designed, auditable measurement experiment rather than as a preliminary inconvenience. The central move is to refuse the question “what is the true search volume?”, which the GT system does not permit us to answer, and to replace it with two questions that are answerable under stated assumptions:

1. Under a fixed query specification and a fixed, preregistered retrieval design, what is the expected value that GT returns for each historical date?
2. On a common scale, what is the latent relative search signal that the repeated retrievals have in common?

The first estimand is design-based and requires no model of Google’s internal sampling; it is unbiased by construction provided the inclusion rule does not depend on the realized values. The second is model-based and requires the assumptions we make explicit in section 3. Reporting both, and keeping them distinct, is a deliberate methodological choice: the design-based quantity is a transparent benchmark against which the model-based reconstruction can be judged.

1.2 Contributions

1. A preregistered acquisition design. Collection day, collection stream, technical replicate, chunk order, and time slot are balanced and locked before any data are seen, and the entire configuration is committed to a SHA-256 protocol hash (section 4). This makes retrieval-side researcher degrees of freedom auditable rather than invisible.
2. Global overlap-graph calibration. We replace sequential stitching with a single weighted least-squares problem on the graph whose nodes are chunks and whose edges are robustly estimated log-scale differences (section 5). This uses all overlap information at once, removes error accumulation, and yields identification, connectivity, conditioning, and cycle-consistency diagnostics that sequential stitching cannot provide.
3. Dependence accounting rather than deduplication. Exact duplicates are retained in the audit archive and collapsed analytically; near-duplicates are identified from residuals after the common signal is removed, so that highly dependent retrievals are not counted as independent evidence (section 6).

4. A three-facet generalizability decomposition. We estimate variance components for historical date, collection day, and collection stream and their interactions, separately for the level, the detection indicator, and the daily innovation (section 7). The associated D-study answers a question practitioners actually face: whether to buy reliability with more collection days, more streams, or more technical replicates.
5. A covariance-adjusted consensus with an honest effective sample size. Weights solve a minimum-variance problem under the estimated residual covariance, and we report spectral and Kish effective pull counts so that dependence is visible in the reported precision (section 9).
6. Propagation into downstream inference. The measurement uncertainty is carried into reliability-corrected OLS and SIMEX rather than discarded at the boundary of the measurement exercise (section 12).

1.3 Scope and what is deliberately not claimed

RACER-GT estimates a latent *relative* search signal on a common scale. It does not estimate absolute Google search counts, and no result in this paper should be read as identifying them. All unbiasedness, consistency, and minimum-variance statements are conditional on the assumptions listed in section 3, and we state in section 14 exactly which guarantees survive which assumption failures.

2 The measurement problem

2.1 Notation

Let $t \in \mathcal{T} = \{1, \dots, T\}$ index historical calendar dates, and let a *complete pull* i denote one full reconstruction of the historical window under the locked query specification. A complete pull consists of J fixed, overlapping chunks; chunk j covers a contiguous window $W_j \subset \mathcal{T}$, and consecutive chunks overlap in at least $m_{\min}$ days.

Write Y_{ijt} for the raw GT value returned for date t in chunk j of pull i . Each pull is characterized by a design cell: a collection day $d \in \mathcal{D}$, a collection stream $s \in \mathcal{S}$, and a technical replicate $r \in \mathcal{R}$.

Remark 2.1 (Streams are composite environments). *A stream is a fixed combination of device, browser profile, cookie state, account state, network route, and IP address. An estimated stream effect is therefore not an IP effect. Attributing it to the IP address alone would require a crossover factorial design that varies IP while holding the remaining factors fixed. We are explicit about this because the applied literature sometimes treats distinct IP addresses as independent sampling units, which the design does not license.*

2.2 Three failure modes of naive construction

Let X_t denote the latent relative search signal we wish to recover.

(F1) Compounding scale error. Under sequential stitching with estimated log-scale corrections $\hat{\ell}_j$, the calibrated value of chunk J carries the error $\sum_{j=2}^J (\hat{\ell}_j - \ell_j)$. Even if each join is unbiased, the variance of the accumulated error grows with the number of joins, so the oldest or newest part of a long series is systematically less reliable than the middle.

(F2) Dependence masquerading as replication. If N retrievals share a common retrieval-side disturbance, the variance of their mean does not fall as $1/N$. Reporting N as a sample size overstates precision. In the extreme case of exact duplicates, N retrievals contain the information of one.

(F3) Attenuation downstream. If the constructed index enters a regression as a covariate measured with classical additive error of variance σ_u^2 , the OLS slope is attenuated by the reliability ratio $\lambda = \sigma_W^2 / (\sigma_W^2 + \sigma_u^2)$. Measurement work that stops at the construction stage leaves this bias in place.

3 Estimands and identification

3.1 Two estimands

Definition 3.1 (Retrieval-mixture expectation). *Fix the query specification Q and a retrieval design that places known probability π_h on design cell $h \in \mathcal{H}$. The retrieval-*

mixture expectation is

$$\theta_t^\pi = \sum_{h \in \mathcal{H}} \pi_h \mathbb{E}[Y_t \mid Q, h], \quad (1)$$

where the inner expectation is taken over the GT system's own randomness conditional on the cell.

Definition 3.2 (Latent common-scale signal). *Let Z_{it} denote the calibrated value of pull i at date t after within-pull scale alignment and baseline normalization. The latent common-scale signal X_t is defined by the measurement model*

$$\mathbf{Z}_t = \mathbf{1}X_t + \mathbf{e}_t, \quad \mathbb{E}[\mathbf{e}_t] = 0, \quad \text{Var}(\mathbf{e}_t) = \Sigma. \quad (2)$$

Definition 3.1 requires no model of the retrieval error and is estimable by stratified averaging. Definition 3.2 requires the assumptions below but supports variance-optimal estimation.

3.2 Assumptions

Assumption 3.1 (Protocol invariance). *Every retrieval in a batch uses the identical query specification, historical endpoints, and chunk manifest, and the configuration hash is unchanged across the batch.*

Assumption 3.2 (Value-independent inclusion). *Whether a retrieval is included in the analysis does not depend on the values it returned. Retrievals may be excluded for documented operational reasons recorded before inspection of the values.*

Assumption 3.3 (Positive-overlap multiplicative model). *For a date t present in chunks j and k of the same pull, with both raw values at or above a prespecified threshold $c_{\min}$,*

$$Y_{jt} = a_j X_t \exp(\epsilon_{jt}), \quad a_j > 0, \quad (3)$$

so that with $\ell_j = \log a_j$ and $\nu_{jkt} = \epsilon_{jt} - \epsilon_{kt}$,

$$r_{jkt} = \log Y_{jt} - \log Y_{kt} = \ell_j - \ell_k + \nu_{jkt}. \quad (4)$$

Assumption 3.4 (Connected overlap graph). *The graph $G = (V, E)$ with V the set of chunks and E the set of chunk pairs having at least $m_{\min}$ usable overlapping days is connected.*

Assumption 3.5 (Conditionally centred edge error). $\mathbb{E}[\nu_{jkt} \mid (j, k) \in E] = 0$ *for every*

edge, and the edge-level estimator is applied to a set of overlap days selected without reference to the realized differences.

Assumption 3.6 (Common signal with centred retrieval error). *Model (2) holds with Σ finite and positive definite on the retained pulls.*

Remark 3.1 (Where each assumption can fail). *Assumption 3.3 fails near zero, because rounding makes the log ratio explode; this is why only cells at or above $c_{\min}$ contribute to edges, while zeros are retained in the final series and analyzed separately through detection reliability. Assumption 3.4 fails if the chunk design leaves a gap; the implementation raises an error rather than silently returning a partially identified series. Assumption 3.6 fails if a subset of pulls shares a retrieval-side bias that is not shared by the rest; this is precisely what the near-duplicate diagnostics in section 6 are designed to expose.*

4 Stage 0: preregistered acquisition design

The design fixes, before collection begins: the query specification; the historical endpoints; the chunk manifest (window length, step, minimum overlap); the set of collection-day offsets; the set of streams; the number of technical replicates; the time slots; and the acceptance thresholds. Stream order is rotated across collection days and chunk order is rotated across pulls, so that order effects are balanced rather than confounded with the facets of interest.

The entire configuration is serialized canonically and hashed:

$$h = \text{SHA256}\{\text{canonical}(Q, \mathcal{D}, \mathcal{C}, \mathcal{T})\}, \quad (5)$$

where $\mathcal{C}$ collects the calibration settings and $\mathcal{T}$ the decision thresholds. Every raw observation may store h . Any change to a locked setting changes h , so configuration drift between the design and the analysis is detectable rather than assumed away.

Pilot separation. An initial monitoring window may be used to check for gaps, query drift, exact duplicates, and connectivity. If that inspection leads to any change in thresholds, stitching rules, query, code, or decision tree, the monitored data must be relabelled as pilot and the confirmatory batch restarted. Otherwise the same data are used both to design and to validate the acceptance rule, producing data-dependent acceptance bias. For the same reason, all GT processing rules

must be locked before any return, volatility, spread, or VaR result is examined; selecting the construction that gives the most significant financial result converts measurement validation into outcome-guided model selection.

5 Stage 1: global overlap-graph calibration

5.1 Robust edge estimation

For each chunk pair (j, k) with at least $m_{\min}$ usable overlap days, the observed log ratios $\{r_{jkt}\}$ are summarized by a Huber M-estimator of location (Huber, 1964), which solves

$$\sum_t \psi_c \left(\frac{r_{jkt} - \mu}{\hat{\sigma}} \right) = 0, \quad \psi_c(u) = \max\{-c, \min(c, u)\}, \quad (6)$$

with $c = 1.345$ and $\hat{\sigma}$ a robust scale. The estimator returns the location $\hat{r}_{jk}$, a robust scale, and a variance $\hat{v}_{jk}$ that becomes the edge precision. Robustness matters here because a small number of overlap days can be contaminated by revisions or by the rounding boundary, and a sample mean of log ratios is not resistant to those points.

5.2 Graph weighted least squares

Collect the edges into a design matrix B with rows indexed by edges: row e for edge (j, k) has $+1$ in column j , -1 in column k , and zero elsewhere. Because only differences are identified, one chunk is fixed as the reference by setting $\ell_{\text{ref}} = 0$; the implementation selects the highest-degree node, which minimizes the leverage of any single poorly estimated edge. With $\mathbf{r}$ the vector of edge estimates and $\Omega = \text{diag}(\hat{v}_e)$,

$$\hat{\ell} = \arg \min_{\ell} (\mathbf{r} - B\ell)^\top \Omega^{-1} (\mathbf{r} - B\ell) = (B^\top \Omega^{-1} B)^{-1} B^\top \Omega^{-1} \mathbf{r}. \quad (7)$$

Proposition 5.1 (Identification). *Under assumption 3.4, B has rank $|V| - 1$ after the reference normalization, so $\hat{\ell}$ in (7) is uniquely defined.*

Proof. B is the incidence matrix of G . For a graph with $|V|$ nodes, the rank of the incidence matrix equals $|V|$ minus the number of connected components. Connectivity gives one component, hence rank $|V| - 1$. Deleting the reference column re-

moves exactly the one-dimensional null space spanned by $\mathbf{1}$, leaving a full-column-rank matrix and a positive definite normal matrix $B^\top \Omega^{-1} B$. $\square$

Proposition 5.2 (Conditional unbiasedness and efficiency). *Under ?? 3.3–3.5, and treating Ω as known, $\hat{\ell}$ is unbiased for ℓ and is the minimum-variance estimator within the class of linear unbiased estimators of ℓ based on the edge estimates.*

Proof. Write $\mathbf{r} = B\ell + \boldsymbol{\nu}$ with $\mathbb{E}[\boldsymbol{\nu}] = 0$ by assumption 3.5 and $\text{Var}(\boldsymbol{\nu}) = \Omega$. This is a linear model with full-column-rank design after normalization, so the Gauss–Markov theorem applies directly to the weighted least-squares estimator (7). $\square$

Remark 5.1 (Why this dominates sequential stitching). *Sequential stitching is the special case in which only the $J - 1$ consecutive edges are used and each is applied in turn. Proposition 5.2 uses every edge, including the non-consecutive overlaps that a long window and a short step generate, and weights each by its estimated precision. It also makes error accumulation impossible: no estimate is defined as a function of a previously calibrated estimate. In addition, the residual $\mathbf{r} - B\hat{\ell}$ has a direct interpretation as cycle inconsistency, which is reported as a diagnostic and has no analogue in the sequential procedure.*

5.3 Reconstruction and normalization

Calibrated observations are $\tilde{Y}_{jt} = Y_{jt} \exp(-\hat{\ell}_j)$, optionally with a lognormal bias correction. Overlapping dates are aggregated by inverse-variance weighting. The resulting series is normalized so that its mean over the prespecified baseline window equals 100. This normalization defines the scale of the estimand; it is not an estimate of absolute search volume.

6 Stage 2: duplicate and dependence diagnostics

Two retrievals may be numerically identical, or nearly so. Both cases violate the implicit assumption that repeated pulls contribute independent information, but they require different treatment.

Exact duplicates. Vectors are hashed after rounding to a fixed number of decimals. Exact groups are retained in the audit archive—deleting them would destroy evidence about the retrieval process—and are collapsed analytically before the consensus step, with multiplicity recorded.

Near-duplicates. Raw correlation between two pulls is uninformative, because all pulls share the same strong latent signal and are therefore correlated at close to one by construction. The diagnostic instead removes the common signal first and examines the residual correlation, together with value-anchored measures: mean absolute deviation on the 0–100 scale, exact cell agreement, and the Jaccard index of the positive-value sets. Pairs that exceed the prespecified thresholds form the edges of a dependence graph.

Remark 6.1 (Components are diagnostic, not equivalence classes). Connected components of the dependence graph are reported as connectivity sets. Membership in a component does not imply that every pair inside it exceeds the threshold, because connectivity is transitive while the pairwise rule is not. The acceptance rule uses the share of the batch occupied by the largest component, which is the quantity that actually matters for whether the batch is dominated by one dependence cluster.

7 Stage 3: generalizability decomposition

Classical reliability coefficients assign all non-signal variation to a single undifferentiated error term. Generalizability Theory (Cronbach et al., 1972; Brennan, 2001) instead decomposes it across the facets of the measurement design, which is what a researcher needs in order to decide where to spend the next unit of collection effort.

For a fully crossed design over historical date t , collection day d , stream s , and replicate r ,

$$Y_{tdsr} = \mu + T_t + D_d + S_s + (TD)_{td} + (TS)_{ts} + (DS)_{ds} + (TDS)_{tds} + E_{tdsr}. \quad (8)$$

Variance components are estimated by the method of moments from the balanced ANOVA sums of squares. The implementation validates full crossing and equal cell counts before estimating, and raises an error otherwise, rather than silently producing components from an unbalanced design.

Remark 7.1 (Confounding with a single replicate). When $n_r = 1$, E cannot be separated from (TDS) ; the implementation retains the combined term and reports it as such. This is stated rather than hidden because it changes the interpretation of the residual component.

The decomposition is run on three transformations, which answer three differ-

ent questions:

- level, Y itself: do relative magnitudes reproduce?
- detection, $\mathbb{I}\{Y > 0\}$: does the zero/non-zero pattern reproduce? This is the binding constraint for low-volume terms.
- innovation, $\Delta \log(1 + Y)$: do daily *changes* reproduce? This is the relevant question when the index enters a model in differences, and it is systematically harder to satisfy than the level criterion.

The generalizability coefficient for relative decisions and the dependability coefficient for absolute decisions follow in the usual way, and the D-study evaluates them on a grid of hypothetical designs, so the marginal value of an additional collection day can be compared with that of an additional stream.

8 Stage 4: the design-based reference estimator

Let h index the prespecified collection-day $\times$ stream cells with target weights π_h summing to one, and let n_h be the number of completed pulls observed in cell h . The stratified estimator is

$$\hat{\theta}_t^\pi = \sum_h \pi_h \frac{1}{n_h} \sum_{i \in h} Z_{it}. \quad (9)$$

Proposition 8.1 (Design unbiasedness). *Under ?? 3.1?? 3.2, and provided $n_h \geq 1$ for every cell with $\pi_h > 0$, $\hat{\theta}_t^\pi$ is unbiased for the retrieval-mixture expectation θ_t^π of definition 3.1.*

Proof. Conditional on cell h , the retrievals in that cell are draws from the GT system under a fixed specification, so $\mathbb{E}[n_h^{-1} \sum_{i \in h} Z_{it}] = \mathbb{E}[Y_t \mid Q, h]$ by assumption 3.2, which guarantees that inclusion carries no information about the values. Taking the π -weighted sum over cells and applying linearity gives (1). $\square$

Remark 8.1 (Dependence affects the variance, not the bias). *Proposition 8.1 does not require the retrievals to be independent. Dependence across pulls inflates $\text{Var}(\hat{\theta}_t^\pi)$ but leaves the expectation unchanged. This is exactly why the estimator is retained as a transparent reference: its validity rests on the design, not on a covariance model that could itself be misspecified.*

9 Stage 5: covariance-adjusted consensus

9.1 Minimum-variance weights

Under assumption 3.6, consider linear estimators $\hat{X}_t = \mathbf{w}^\top \mathbf{Z}_t$ with $\mathbf{w}^\top \mathbf{1} = 1$, which is the condition for unbiasedness.

Proposition 9.1 (BLUE weights). *Under assumption 3.6 with Σ known and positive definite,*

$$\mathbf{w}^* = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}} \quad (10)$$

solves $\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w}$ subject to $\mathbf{w}^\top \mathbf{1} = 1$, and the resulting $\hat{X}_t = \mathbf{w}^{\top} \mathbf{Z}_t$ has variance $(\mathbf{1}^\top \Sigma^{-1} \mathbf{1})^{-1}$, the minimum over all linear unbiased estimators.*

Proof. The Lagrangian is $L = \mathbf{w}^\top \Sigma \mathbf{w} - 2\lambda(\mathbf{w}^\top \mathbf{1} - 1)$ with first-order condition $\Sigma \mathbf{w} = \lambda \mathbf{1}$, hence $\mathbf{w} = \lambda \Sigma^{-1} \mathbf{1}$. Imposing $\mathbf{w}^\top \mathbf{1} = 1$ gives $\lambda = (\mathbf{1}^\top \Sigma^{-1} \mathbf{1})^{-1}$ and (10). The objective is strictly convex because $\Sigma \succ 0$, so the stationary point is the unique minimum. Substituting yields the stated variance. $\square$

9.2 Stabilization in finite samples

Proposition 9.1 assumes Σ known. In practice the number of pulls is of the same order as, or smaller than, the number of dates used to estimate the covariance, so the sample covariance is ill-conditioned and its inverse is unstable. The implementation therefore:

1. collapses exact duplicates analytically, so that a repeated vector cannot inflate the apparent rank;
2. removes a pull-level offset, so that a constant retrieval bias is not mistaken for signal disagreement;
3. estimates Σ by Ledoit–Wolf shrinkage (Ledoit and Wolf, 2004) toward a well-conditioned target;
4. imposes $w_i \geq 0$ and $w_i \leq \bar{w}$ while retaining $\sum_i w_i = 1$.

Remark 9.1 (What the constraints cost). *The nonnegativity and cap constraints are stabilization devices, not improvements on proposition 9.1. The strict BLUE property belongs to the unconstrained solution. The constrained solution is a minimum-variance convex combination: it gives up the theoretical optimum in exchange for*

refusing the large offsetting positive and negative weights that an ill-conditioned $\hat{\Sigma}^{-1}$ would otherwise produce. We regard this as the right trade for a measurement instrument that must behave predictably on data the analyst has not yet seen.

9.3 Effective number of pulls

Because dependence is the central concern, the raw count of pulls is not reported as a sample size. We report instead the Kish effective count (Kish, 1965) $1/\sum_i w_i^2$ and a spectral effective count derived from the eigenvalues of the residual correlation matrix. Both fall below the nominal count when retrievals are dependent, and the acceptance rule in section 11 is stated in terms of the spectral count.

10 Stage 6: temporal benchmarking

Long GT histories at daily frequency are not directly comparable to the weekly or monthly series that GT returns for the same query over the full window. Let z be the preliminary daily consensus, A the aggregation matrix mapping days to benchmark periods, and b the lower-frequency benchmark. The soft benchmark solves

$$\min_x (x - z)^\top Q (x - z) + (Ax - b)^\top W (Ax - b), \quad (11)$$

where Q combines fidelity to the daily consensus with a smoothness penalty on the correction path, and W is the benchmark precision. This is the quadratic-programming form of the movement-preservation idea familiar from Denton (1971) and Chow and Lin (1971); treating b as noisy rather than exact is appropriate here because the lower-frequency GT series is itself a sampled quantity.

Proposition 10.1 (Existence and uniqueness). *If $Q \succ 0$ and $W \succeq 0$, the objective in (11) is strictly convex and the solution is the unique $x^* = (Q + A^\top W A)^{-1}(Qz + A^\top W b)$.*

Proof. The Hessian is $2(Q + A^\top W A)$, which is positive definite because $Q \succ 0$ and $A^\top W A \succeq 0$. Setting the gradient to zero gives the stated normal equations, whose coefficient matrix is invertible. $\square$

11 Stage 7: reliability and the acceptance decision tree

Three reliability questions are kept separate because they can have different answers for the same series: whether the zero/non-zero pattern reproduces (detection, summarized by Fleiss’ κ , Fleiss, 1971), whether levels reproduce, and whether daily innovations reproduce.

The acceptance decision is a locked tree evaluated after estimation. Each rule compares an observed diagnostic against a prespecified threshold and returns pass, fail, or indeterminate. The batch status is

- FAIL if any mandatory rule fails;
- REVIEW if no mandatory rule fails but at least one is indeterminate;
- PASS otherwise.

The rules cover protocol integrity, overlap-graph connectivity, the number of numerically distinct pulls, the spectral effective pull count, the share of zero observations, level and innovation generalizability, detection agreement, concentration of the near-duplicate dependence graph, consensus convergence as the last pull is added, and consistency with the lower-frequency benchmark.

Remark 11.1 (Indeterminate is not a soft fail). The detection rule is not identified when every observation falls in one detection category, because κ is undefined without variation. The implementation reports the rule, marks it indeterminate, and—crucially—makes it non-mandatory in exactly that case, so that a series with no zeros is not penalized for a coefficient that cannot exist. Conflating “cannot be computed” with “failed” would systematically reject high-volume terms.

Remark 11.2 (What PASS does and does not mean). PASS authorizes construction under the locked protocol. It does not establish that the keyword has construct validity for the economic concept of interest, it does not identify absolute search counts, and it licenses no causal interpretation.

12 Stage 8: propagating measurement error downstream

Let $W_t = X_t + u_t$ be the constructed index, used as a regressor for an outcome Y_t with controls. Under classical additive measurement error, the moment-corrected estimator is

$$\hat{\beta} = \frac{\mathbf{W}^\top \mathbf{M} \mathbf{Y}}{\mathbf{W}^\top \mathbf{M} \mathbf{W} - \text{tr}(\mathbf{M} \Omega_u)}, \quad (12)$$

where M is the residual maker for the controls and Ω_u the measurement error covariance supplied by the construction stage. Standard errors use a HAC estimator (Newey and West, 1987). The implementation raises an error when the corrected denominator is non-positive, rather than returning an explosive or sign-flipped coefficient; a non-positive denominator means the estimated measurement error exceeds the observed regressor variation, which is a statement about the data, not a numerical detail to be smoothed over.

As an alternative that does not require the classical structure to hold exactly, SIMEX (Cook and Stefanski, 1994; Carroll et al., 2006) adds further known noise on a grid of λ values, records the resulting estimates, fits an extrapolation function, and evaluates it at $\lambda = -1$.

13 Controlled Monte Carlo evidence

The simulation generates a latent signal with realistic dynamics, applies collection-day and stream effects, adds correlated retrieval error, splits the history into independently normalized chunks, imposes rounding and zeros, and optionally injects exact duplicates. Because the latent truth is known by construction, estimators can be compared directly.

Table 1: Twenty controlled replications. Lower RMSE is better.

Estimator	Mean RMSE	Mean corr.	Mean innov. corr.	Mean peak recall
Single pull	7.2305	0.9695	0.8869	0.8868
Cross-pull median	4.1710	0.9898	0.9556	0.9447
Simple mean	3.7253	0.9919	0.9747	0.9526
RACER-GT	3.5265	0.9927	0.9745	0.9526

Under this data-generating process, RACER-GT reduces mean RMSE by approximately 51.2% relative to a single pull and by 5.3% relative to the simple mean.

Remark 13.1 (Reporting the losses as well as the wins). RACER-GT does not dominate the simple mean on every statistic in table 1: mean innovation correlation is marginally lower (0.9745 versus 0.9747) and peak recall is identical. We report this because the honest summary is that the gain concentrates in level accuracy, where scale-error correction and dependence-aware weighting have most to contribute, and is negligible for differenced quantities under this particular DGP. A practitioner whose model uses only innovations should weigh that accordingly.

14 Scope of the statistical guarantees

We state precisely what is and is not established.

1. Under ?? 3.1?? 3.2, the stratified design-cell average is design-unbiased for the prespecified retrieval-mixture expectation (proposition 8.1). This holds *without* any independence assumption across pulls.
2. Under ?? 3.3–3.5, the relative log-scale parameters are identified and the graph WLS estimator is conditionally unbiased and efficient among linear estimators built on the edge estimates (propositions 5.1 and 5.2).
3. Under assumption 3.6 with Σ known or consistently estimated, the unconstrained GLS weights are minimum-variance among linear unbiased estimators (proposition 9.1). The constrained default is a stabilized convex combination and does not inherit this optimality.
4. Given a correct lower-frequency benchmark and an identified constraint set, the benchmark estimator is the unique minimizer of an explicit quadratic objective (proposition 10.1).
5. Under classical additive measurement error with a consistent estimate of Ω_u , (12) corrects attenuation bias.

None of these results establishes unbiasedness with respect to Google’s absolute search counts, and none is claimed to hold unconditionally.

14.1 Known limitations

- The variance components use balanced method-of-moments estimation. REML for unbalanced designs is a planned extension.
- Consensus confidence intervals are conditional approximations. A full-pipeline moving-block bootstrap—resampling historical blocks and re-estimating edges, scales, duplicates, covariance, weights, and the benchmark before re-estimating the financial model—is the appropriate procedure for confirmatory inference.
- Benchmark noise, scale-calibration uncertainty, and tuning-parameter uncertainty are not yet fully propagated into the reported intervals.
- Zero values may mix genuine zero search, privacy thresholds, sampling, and rounding. Version 1.0.0 retains zeros and isolates detection reliability, but

does not implement an interval-censored latent-count likelihood. For high zero-share terms, moving to weekly frequency or to a broader Topic is usually more defensible than manufacturing daily values.

- Anchor-bank calibration across different queries is documented as an extension but is not implemented in version 1.0.0.
- The package is deliberately ingestion-first and embeds no unofficial scraper, because endpoints, rate limits, authentication, and terms of service change. Collectors need only emit the documented long-format schema.

15 Conclusion

Google Trends is a measurement instrument with a documented tendency to return different numbers for the same question. Treating its output as an error-free observation is not a conservative simplification; it silently imports scale error, dependence, and attenuation into whatever model consumes it.

RACER-GT takes the opposite stance. It fixes the retrieval design before any data are seen, calibrates all chunks simultaneously on the overlap graph, measures rather than assumes the independence of repeated retrievals, decomposes the remaining variation across the facets of the design, and carries the resulting uncertainty forward into downstream estimation. What it delivers is not certainty about Google’s internal counts, which is unavailable, but a reproducible, auditable, and explicitly conditional estimate of a well-defined target, together with a locked rule for deciding whether a given batch of data is fit to use at all.

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